DETECTION_LAB_REQUIREMENTS

Detection Lab is a Flyreel-internal tool for building, evaluating, and refining VLM-based visual detection prompts (e.g. roof corrosion, missing siding, hazard identification). It supports the full lifecycle of a detection: building labeled image datasets, annotating them through a structured QA pipeline, drafting and iterating prompts against those datasets, comparing prompt versions, and running held-out evaluations before promoting a detection to production.

The current POC runs locally on SQLite with file-system storage and no authentication. This document captures the requirements to harden the POC into a hosted, multi-user, role-gated platform tool integrated alongside the existing Flyreel dashboard. The work is split into three milestones, each delivering a coherent MVP:

- **Milestone 1 — Annotation MVP:** General platform integration + Datasets, Annotation, and QA tabs. Delivers a production-grade labeling pipeline that produces auditable, finalized datasets.
- **Milestone 2 — Prompt Building MVP:** Detection Setup, Build & Run Datasets, HiL Review, Detections & Logs, and Admin tabs. Delivers the core prompt authoring loop on top of the finalized datasets from M1.
- **Milestone 3 — Prompt Enhancement & Evaluation MVP:** Prompt Feedback, Prompt Compare, and Held-Out Eval tabs. Delivers automated improvement suggestions, side-by-side prompt comparison, and the final regression gate before deploying a detection.

MILESTONE 1 — Annotation MVP

Milestone 1 delivers a production-grade labeling pipeline: annotators can be assigned datasets, label them image-by-image, flag uncertain items, and QA managers can sample, resolve flags, finalize datasets, and track annotator performance — all under platform auth and RBAC. After M1, Detection Lab is a usable annotation tool even before any prompt-building features are migrated.

The three tabs in scope for M1 are **Saved Datasets**, **Annotation**, and **QA**. The prompt-related tabs are intentionally deferred to M2 so M1 can ship without depending on inference infrastructure.

Section 1: General App Requirements

This section covers cross-cutting platform-integration requirements that are independent of any single tab. It expands the requirements table the user already drafted (authentication, RBAC, data persistence, hosting, storage, security).

Image ID Uniqueness

Image IDs must be globally unique across the platform, with one exception: a **parent dataset and any datasets linked to it as children** (e.g. duplicate-annotation copies created for QA discrepancy review) share the same image IDs by design, so that label disagreements can be computed image-for-image. Uploading an image whose ID already exists outside of a parent-child link should be rejected at the API layer with a clear error.

Annotator Identity & Impersonation

- The annotator selector dropdown (visible in the Annotation tab and in any "assign to annotator" modal) is populated from the membership of the **Annotator** user group in the platform — not from a hand-maintained list.
- When a user logged in as **Annotator** opens the Annotation tab, they only ever see their own dashboard; the annotator selector is hidden or locked to their own identity.
- When a user logged in as **Super Admin** opens the Annotation tab, they can use the annotator selector to **impersonate** any annotator and see exactly what that annotator's dashboard looks like (assigned datasets, flags, performance). Impersonation is read-through (Super Admin actions taken under impersonation are logged as Super Admin, not as the impersonated annotator).

Category	Requirement	User Story	Details	Acceptance Criteria
Authentication	*1.1 – Enforced Authentication*	As a user, I must log in before accessing	All users must authenticate before	Unauthenticated users are redirected to login; all API

		the tool so that access is controlled.	accessing any part of the application. No anonymous access permitted.	requests without valid auth return 401.
	1.2 – Session Management	As a platform admin, I want user sessions to be secure and expire appropriately.	Secure session handling with expiration and logout invalidation.	Sessions expire after configured time (same as prod platform timeout); logout invalidates session token immediately.
Authorization	1.3 – Creation of a New Data Annotator User Group		New platform user group called Data Annotator with view and edit access to ***only*** the Annotation tab in Detection Lab.	
	1.4 – Role-Based Access Control	As an admin, I want to control who can access and modify data.	System must support Super Admin and a new Data Annotator **Super Admin** • View and edit access to all tabs and Actions **Data Annotator** • View and edit access to the Annotation tab.	*API-Level Enforcement* – Role assignments persist; unauthorized actions return 403. *UI Permission Enforcement* – Tabs are hidden/disabled based on role.
Data	*1.1 – Shared Workspace Model*	As a user, I want to see the same data as my team.	All detections, prompts, datasets, and runs are globally visible to authorized users.	Multiple users see identical datasets and results across sessions/devices.
	1.2 – Cross-Session Persistence	As a user, I want my work to persist and be accessible later.	Data persists across logins and devices.	Data created by User A is visible to User B after login.
	1.3 – Concurrency Handling	As a user, I want to avoid losing work due to concurrent edits.	System must handle concurrent edits without corruption.	Simultaneous updates do not overwrite silently (e.g., optimistic locking or last-write-wins defined).
Backend	*1.4 – Migration to Production DB*	As a system, I need a scalable database for multiple users.	Replace SQLite with PostgreSQL.	Application runs with external DB; no local DB dependency remains.
Storage	*1.5 – Cloud Object Storage*	As a system, I need durable storage for datasets.	Replace local file storage with GCS or S3.	Uploaded files persist across deployments and instances.
	1.6 – Secure File Access	As a user, I want secure access to	Files must not be publicly exposed; use	Direct public access is blocked; access requires

		uploaded data.	signed URLs.	signed URLs.
Infra	*1.7 – Hosted Deployment*	As a user, I access the tool without running it locally.	System must run as a hosted service, not local-only.	App accessible via URL; not dependent on localhost runtime.
	1.8 – Stateless Services	As a system, I scale horizontally.	Application instances must be stateless.	Multiple instances can run without data inconsistency.
Security	*1.9 – Authenticated APIs*	As a system, I must secure all endpoints.	All API endpoints require authentication.	All unauthenticated requests return 401.
	1.10 – Rate Limiting	As a system, I prevent abuse and overload.	Rate limiting must be enabled for write-heavy endpoints, BUT it also needs to be able to support runs of 1000 images hitting Gemini endpoint.	Excess requests are throttled or rejected. Can process runs of up to 600 images in sequence.
	1.11 – Formal Security Review	As a company, we ensure the tool meets security standards.	System must pass internal security review before production.	Review completed and approved by security stakeholders.

Section 2: Saved Datasets Tab (M1)

Purpose

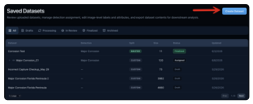
Saved Datasets is the centralized hub for managing all image collections in Detection Lab. It is where datasets are inspected, edited, assigned to annotators, linked together for duplicate-annotation QA, appended to, exported, and deleted. Every other tab that needs a dataset reads it from here.

Workflow

1. A super admin user opens the Saved Datasets tab and either:
   a. Builds a new dataset by naming it, optionally assigning it to a detection, selecting a split type (MASTER, TRAIN, TEST, EVALUATE, CUSTOM), and uploading images (files / CSV); or
   b. Opens an existing dataset to inspect, edit metadata, append more images, or assign it to annotators and create linked child datasets.
2. The user assigns the dataset to one or more annotators from the platform Annotator group. Assignment creates a copy of the dataset (with optional label/attribute reset) for each assigned user and routes the assigned dataset to that annotator.
3. When QA later approves the separate child datasets after performing QA and label/attribute discrepancy reconciliation, the child datasets are then merged back into the parent dataset, shifting it's status to finalized and making available as input to runs (M2) and held-out evals (M3). The child datasets are then archived.

Reference screenshots

Requirement	Details	User Story	Notes	Designs

1.12 – Build New Dataset	"Create Dataset" action opens a build flow supporting two upload methods: 1. Drag-and-drop image files 2. CSV upload with columns imageId, imageURL, groundTruthLabel, attributes. After upload, the user names the dataset, picks a split type, optionally assigned it to a detection and saves.	As a system admin, I want to spin up a new dataset from a folder of images or a CSV created from a Metabase export of image URLs.	Image IDs must be globally unique on upload. Reject duplicates with a clear error pointing at the conflicting dataset. When CSVs are uploaded, they must have the following columns: • imageId • imageUrl • groundTruthLabel • attributes **Include ability to download sample CSV: `dataset-manifest-example.csv` Image ID and URL have to be populated, but groundTruthLabel and attributes can be left blank. Support the following dataset split types: • Master • Train • Test • Evaluate • Custom *Changes:* • Upload Dataset button renamed to "Create Dataset" • Removed the JSON upload option, as it is superfluous.	 Select Create Dataset  Image files  CSV manifest Pressing Upload Dataset once files are attached creates the datatset and closes out the create dataset modal.
1.13 – Dataset Table	Top-level list of all datasets. Columns: • Name • Detection • Split Type (badge) • Size (item count), • Status (badge) • Updated (date)	As a user, I want to find and triage datasets quickly.	Support for linked datasets in the Datasets Table. Statuses: • Drafts • Assigned • In Annotation • Submitted • In QA	

	Sortable on Name, Size, Created, Updated. Full-width tab filter above the table: • All • Drafts • Processing • In Review • Finalized • Archived		• Needs Revision • Approved • Finalized • Archived For the simplified full width tab filter above the table, • Drafts = all datasets in draft status. • Processing = all datasets in assigned, in annotation, and needs revision • In Review = all datasets in submitted, In QA, and approved status • Finalized = all datasets in finalized status • Archived = all datasets in archived status	
1.15 – Expandable Row → Details Panel	Clicking a row expands a details panel below the table showing: basic metadata (name, detection, split type, size, QA status, assigned annotators, created/updated), and a scrollable item list with columns Image ID, thumbnail, Ground Truth Label, AI Prediction (if any runs have been completed), Segment Tags, Item Status. Item list is filterable by label and status.	As a user, I want to inspect a dataset's contents without leaving this page.	Thumbnails lazy-load. Clicking a thumbnail opens the shared Image Preview Modal.	
Actions menu on Dataset Details table	In the upper right corner of the Dataset Details table, there should be an action menu that supports the following options: • Edit Details • Export CSV • Assign Annotators			

	• Delete Dataset			
Edit Dataset Metadata Modal	Edit name, detection, split type. Auto-split toggle visible only when split type is MASTER (auto-partitions into train/test on save).	As a user, I want to correct metadata after creation.	Changing detection on a non-empty dataset should warn — segment taxonomies may not match.	
Export Dataset	Export button on the dataset details panel. Downloads the dataset as a CSV (one row per image). Includes image IDs, labels, segment tags, image URIs, and (if present) latest AI prediction.	As a Data Analyst, I want to hand off finalized datasets for downstream use.	Available to all roles with view access.	
Assign to Annotators Modal	"Assign to Annotators" opens a modal listing all members of the platform Annotator user group as a checklist. Already-assigned annotators are shown as read-only. Two checkboxes: "Reset their labels before assignment" and "Reset their segment tags before assignment". On assign, a child dataset is created per selected annotator, sharing image IDs with the parent (this is the parent-child linkage that req 4.4 allows).	As a Super Admin, I want to send the same images to multiple annotators for independent labeling.	Annotator list **must** come from the live platform Annotator group (req 3.4) — no hand-maintained list. Assigning creates entries with qa_status = "assigned" visible in the annotator's Annotation tab.	
Annotator Group Drives Selector	Annotator dropdowns are populated from membership in the platform's Annotator user group.	As an admin, I want adding/removing annotators in the platform to automatically reflect in Detection Lab.	New members of the Annotator group appear in dropdowns within one session refresh; removed members disappear.	
Delete Dataset	Delete button visible only to Super Admins (req 3.1). Disabled if the dataset has any associated runs (M2+). Hard delete; surfaces a	As a Super Admin, I want to clean up draft or obsolete datasets.	Deleting a dataset that is the parent of a linked child should either cascade or refuse — define before implementation.	

	confirmation modal listing what will be deleted (items, flags, linked siblings).		Recommended: refuse, force user to unlink first.	
Image Preview Modal	Shared modal across tabs. Full image with zoom (1×–4×) and pan, plus a side panel showing Image ID, URI, ground-truth label, AI prediction (if any), segment tags, and image description/notes.	As a user, I want to inspect an image without leaving the table view.	Same component is reused in Annotation, QA, HiL Review.	
Status Badges	Use a consistent color-coded status set across the app: Draft (gray), Assigned (blue), In Progress (amber), Submitted (purple), In QA (teal), Needs Revision (red), Approved (green), Finalized (deep green), Archived (slate).	As a user, I want to read status at a glance anywhere in the app.	The same badge component must be used in Saved Datasets, Annotation, and QA tabs.	
Empty / Loading States	Empty state when no datasets exist: short copy + "New Dataset" CTA. Skeleton rows while loading.	As a user, I want clear feedback when the table is empty or loading.		

Section 3: Annotation Tab (M1)

Purpose

The Annotation tab is the labeling workspace. It is the **only** tab visible to users in the Annotator role. Annotators see datasets that have been assigned to them, work through them image-by-image to assign ground-truth labels and segment tags, leave notes, and flag images they cannot confidently label. They can also review their own flags and personal performance metrics. Super Admins can use this tab in impersonation mode to QA any annotator's experience.

Workflow

1. The annotator logs in and lands on the Annotation tab → My Datasets sub-tab. They see four summary cards (Assigned / Needs Attention / In Progress / Complete) and a filterable table of their assigned datasets.
2. They click a dataset to enter the image-by-image annotation view. The viewer is a two-column layout: the image on the left with zoom/pan and keyboard navigation, the review panel on the right with the label buttons, segment-tag toggles, notes textarea, and a "Flag for Secondary Review" action.
3. They progress through images using arrow keys or the prev/next buttons. Notes auto-save on blur or navigation. The progress counter updates live.

4. When done, they hit Submit; the dataset moves to submitted status and is no longer editable to them. Submitted datasets become QA's responsibility.
5. They use the Flags sub-tab to see what they've flagged (and how QA resolved it). They use the Performance sub-tab to see their personal accuracy, flag rate, datasets finalized, and items labeled.

Reference screenshots: screenshots/02_annotation_my_datasets.png, screenshots/03_annotation_flags_open.png, screenshots/04_annotation_flags_resolved.png, screenshots/05_annotation_performance.png, screenshot_annotation_view.png, Screenshot 2026-06-03 at 3.16.45 PM.png, Screenshot 2026-06-03 at 3.29.06 PM.png, Screenshot 2026-06-03 at 3.37.27 PM.png, Screenshot 2026-06-03 at 3.38.54 PM.png.

Requirement	Details	User Story	Notes	Mile
Annotator Identity Selector	Right-aligned dropdown in the Annotation header. For Annotator-role users, the dropdown is locked to their own identity (or hidden). For Super Admins, the dropdown lists every member of the platform Annotator group and switching it switches the entire view into impersonation of that annotator.	As a Super Admin, I want to see what an annotator sees without sharing their login.	Must pull from the platform Annotator user group live (req 3.4). Impersonation actions are logged as Super Admin (req 3.5, 9.1).	M1
Super Admin Impersonation	Super Admin users can view the Annotation tab as any annotator from the annotator selector, seeing that annotator's dashboard exactly as the annotator would.	As a Super Admin, I want to QA an annotator's experience without sharing credentials.	Selecting an annotator while logged in as Super Admin shows that annotator's assigned datasets, flags, and performance metrics; any write actions taken while impersonating are logged with the Super Admin's identity.	
My Datasets Sub-Tab — Summary Cards	Four clickable cards across the top: **Assigned** (inbox icon, blue), **Needs Attention** (warning icon, red — counts needs_revision and datasets with open flags), **In Progress** (clock, amber), **Complete** (check, green — counts submitted/in QA/approved/finalized). Clicking a card filters the table.	As an annotator, I want to see my workload at a glance and jump straight to what needs me.		M1
My Datasets Sub-Tab — Filter Tabs	Full-width pill bar below the cards: All / Action Needed / In Progress / Submitted / Done. Click toggles the table filter.	As an annotator, I want to slice my queue		M1

		without re-sorting.		
My Datasets Sub-Tab — Dataset Table	Columns and widths: **Dataset (30%)** — name plus the manager's revision note when status is needs_revision; **Detection (20%)**; **Progress (22%)** — progress bar plus "X/Y labeled (Z%)"; **Status (15%)** — color-coded badge; **Flags (13%)** — open flag count or "—". Default sort puts needs_revision first, then datasets with open flags, then in_progress, then assigned, then everything else. Clicking a row enters the annotation view.	As an annotator, I want the most urgent work to surface without me hunting.		M1
Annotation View — Header	Back button (returns to My Datasets), large dataset name, detection name + progress counter ("Major Corrosion · 431/500 labeled (86%)"), and a read-only badge when the dataset is in a terminal status (submitted, approved, finalized).	As an annotator, I want to know what I'm working on and whether I can still edit it.		M1
Annotation View — Image Panel (Left)	Toolbar: image counter (X/Y — image_id), Copy ID button, Zoom +/–, Reset, Prev/Next. Large viewport (~500 px tall, responsive). Zoom 1×–4× with drag-to-pan when zoomed. Keyboard nav: left/right arrows move images (disabled while a textarea is focused).	As an annotator, I want a fast, low-friction labeling experience.		M1
Annotation View — Secondary Review Card	If the item is not flagged: a "Flag for Secondary Review" button that opens a modal with a reason textarea and an "I'm certain I can't label this" checkbox. If already flagged: amber indicator, reason visible, and (in QA context only) a "Resolve Flag" button.	As an annotator, I want to surface uncertain items to QA without blocking the whole dataset.	The flag modal is the same component as the one in HiL Review (M2).	M1
Annotation View — Ground Truth Label Card	Horizontal button row: DETECTED / NOT_DETECTED / UNSET. Color-coded active state (purple for DETECTED,	As an annotator, I want one-		M1

	accent for NOT_DETECTED, neutral for UNSET). Clicking persists immediately.	click labeling.		
Annotation View — Attributes Card	Only renders when the detection has a segment_taxonomy defined. Multi-select toggle pills, one per taxonomy option (e.g. "major", "minor", "hairline" for corrosion).	As an annotator, I want to capture detection severity/type alongside the label.	Taxonomy is read from the detection record.	M1
Annotation View — Annotator Note Card	Textarea for free-form notes per image. Auto-saves on blur and on navigation away. Stored on DatasetItem.image_description.	As an annotator, I want to leave context that helps the next reviewer.		M1
Annotation View — Resolved Flag History	When an item that was previously flagged has been resolved by QA, show a card with the original reason, QA's resolution action, resolution note, and date.	As an annotator, I want to learn from how QA handled my flags.		M1
Annotation View — QA Report Card	When the dataset's status is approved or finalized, render a card with the dataset's QA acceptance rate (correct samples / total reviewed) and total correction count from QA sampling.	As an annotator, I want to see how my work performed in QA.		M1
Read-Only Mode	When the dataset is submitted, in_qa, approved, or finalized, all editable controls (label buttons, segment toggles, notes textarea, flag button) are disabled and the read-only badge is shown in the header.	As an annotator, I want to be sure I'm not accidentally editing finalized work.		M1
Submit Dataset Action	A "Submit" CTA appears when 100% of items are labeled. On submit, the dataset's qa_status transitions from in_progress to submitted. Confirmation modal first ("This will hand the dataset to QA. You won't be able to edit it after submitting.").	As an annotator, I want a clear handoff moment.		M1

Flags Sub-Tab	Filter pills at top: Open (N) / Resolved (N). Below: list of the annotator's flags. Each row: image thumbnail (10×10 rounded), Image ID (monospace) + Dataset name (separated by middot), flag reason text, and either a resolution badge+note+date (resolved) or "Flagged [date]" + "Click to view" (open). Clicking opens the shared Image Preview Modal with a Flag Reason section highlighted in amber.	As an annotator, I want one place to track every flag I've raised and how it was resolved.	Match screenshots/03_annotation_flags_open.png and 04_annotation_flags_resolved.png.	M1
Performance Sub-Tab	Four summary cards with hover InfoTips: **Accuracy** ("Correct samples ÷ total reviewed samples"), **Flag Rate** ("Open flags ÷ total items assigned"), **Datasets Finalized** ("Datasets that completed full QA process"), **Items Labeled** ("Total images labeled"). Below, a Finalized Datasets table: Dataset, Items, Accuracy (color-coded — green ≥90%, amber ≥75%, red <75%), Corrections, Status. Empty state if no finalized datasets yet.	As an annotator, I want to see how I'm doing without asking my manager.	Match screenshots/05_annotation_performance.png.	M1
Annotator-Role Tab Visibility	Users whose primary platform role is **Annotator** see only the Annotation tab in Detection Lab's navigation. All other tabs are hidden and their API routes return 403 to that role.	As an annotator, I shouldn't see surfaces I can't use.	Reinforces req 3.1, 3.3.	M1

Section 4: QA (Quality Assurance) Tab (M1)

Purpose. The QA tab is the manager-facing counterpart to Annotation. It is where flagged items are resolved, annotator work is sampled for accuracy, redundantly-labeled datasets are compared for discrepancies, and datasets move through the formal QA

pipeline (Draft → Assigned → In Progress → Submitted → In QA → Needs Revision / Approved → Finalized). It also exposes per-annotator performance metrics and a full activity log.

Workflow.

1. The QA manager opens the QA tab to the Pipeline view and sees a Kanban-style board of every dataset by status. Cards show progress, open flag count, assignee, and last updated.
2. The manager uses Flags Queue to triage open flags: each flag opens the Image Preview Modal where they pick a resolution action (label_confirmed / label_corrected / attributes_corrected / image_removed / needs_discussion) and optionally leave a note. Bulk Resolve All / Dismiss All available for triage.
3. The manager uses QA Sampling to spot-check a finalized annotator's work: choose a strategy (Random / Stratified / Flagged / Discrepancy), set a sample size, generate the sample, and review item-by-item. Outcomes update annotator accuracy metrics.
4. For redundantly-labeled (linked) datasets, the manager uses the Discrepancy view to surface images where the two annotators disagreed and decide which label wins.
5. The manager uses Metrics & Logs to view per-annotator scorecards (accuracy, flag rate, label error rate, attribute error rate, discrepancy rate, correction rate) and a chronological activity log of every QA action.
6. Approved datasets land in Finalized, where they're exportable and become inputs to M2's runs.

Reference screenshots: screenshots/06_qa_dashboard.png, screenshots/06_qa_pipeline.png, screenshots/07_qa_flags_queue.png, screenshots/08_qa_metrics_logs.png, Screenshot 2026-06-03 at 3.55.50 PM.png, Screenshot 2026-06-03 at 3.58.16 PM.png, Screenshot 2026-06-03 at 4.21.55 PM.png, Screenshot 2026-06-03 at 4.22.06 PM.png.

Requirement	Details	User Story	Notes	Milestone
Sub-Tab Navigation	Icon-based sub-tab bar across the top: Pipeline / Flags Queue / QA Sampling / Discrepancies / Metrics & Logs / Finalized. Same active-state styling as elsewhere in the app.	As a QA manager, I want one tab with clear sub-sections, not six different tabs.		M1
Pipeline View — Kanban Columns	Columns for each QA status: Draft, Assigned, In Progress, Submitted, In QA, Needs Revision, Approved, Finalized. Each column lists dataset cards. Card content: dataset name, detection name, assignee,	As a QA manager, I want to see where every dataset is in the pipeline.	Match screenshots/06_qa_pipeline.png.	M1

	open-flag badge, progress bar (items_labeled / size), last-updated timestamp. Card is clickable to open dataset details.			
Pipeline View — Filters	Filter controls above the board: by Status (multi-select), Detection (single), Assignee (single from Annotator group).	As a QA manager, I want to narrow the board when we have a lot in flight.	Assignee dropdown sourced from Annotator group (req 3.4).	M1
Pipeline View — Manual Status Transitions	The manager can move a dataset to a different status from the card menu (e.g. "Send back to annotator" → needs_revision, "Approve" → approved, "Finalize" → finalized). Each transition writes an entry to the QA activity log.	As a QA manager, I want explicit control over the pipeline, not just automatic transitions.	Transitions must respect role permissions and log actor (req 9.1).	M1
Flags Queue — Open Flags Section	List of all flags with status=open. Per-page selector (10/25/50, right-aligned). Each row: thumbnail, Image ID + Dataset name, flag reason, date. Clicking opens the Image Preview Modal. Batch "Resolve All" / "Dismiss All" buttons at top with a resolution action dropdown and optional note.	As a QA manager, I want to triage the flag backlog quickly.	Match screenshots/07_qa_flags_queue.png.	M1

Flags Queue — Image Preview Modal (Flag Resolution)	Modal shows full image with zoom/pan, flag reason highlighted in amber, ground-truth label (editable), segment tags (editable), and a Resolution Action selector with options: label_confirmed, label_corrected, attributes_corrected, image_removed, needs_discussion. Optional resolution note. Resolve / Dismiss buttons at bottom.	As a QA manager, I want all context and action in one place.	Reuses the shared Image Preview Modal with flag-specific extensions.	M1
Flags Queue — Resolved Flags Section	Collapsible section below Open Flags. Same row format but read-only and shows the resolution badge. Cannot be re-resolved.	As a QA manager, I want to look back at how a flag was handled without re-opening it.		M1
QA Sampling — Strategy & Sample Generation	Inputs: Dataset selector, Sampling Strategy (Random / Stratified / Flagged / Discrepancy), Sample Size. "Generate Sample" creates QaSample records.	As a QA manager, I want a reproducible way to spot-check annotator quality.	Strategy semantics: Random = uniform; Stratified = balanced by label; Flagged = only flagged items; Discrepancy = only items disagreeing with a linked sibling dataset.	M1
QA Sampling — Review Table	Table of sampled items: thumbnail, image ID, annotator label vs expected label (if discrepancy strategy), action buttons (Accept / Correct Label / Correct Attributes / Mark Both Corrected / Skip), notes field. Reviewing updates	As a QA manager, I want a fast review loop that produces auditable accuracy metrics.	Outcomes feed into Annotator Performance metrics.	M1

	QaSample.status to "reviewed" and sets the outcome. Past samples for this dataset visible below.			
Discrepancies View	Linked-dataset pair selector. On selection, render a table of images where labels disagree between the two annotators' copies: Image ID, Annotator A label, Annotator B label, Tags A, Tags B, mismatch indicator. Clicking opens the Image Preview Modal with both labels shown. Resolution options: pick the winning label, or mark the image for removal. Trend chart of discrepancy rate over time.	As a QA manager, I want to surface and resolve label disagreements between annotators.	Only available when both datasets are linked via linked_dataset_id and share image IDs (req 4.4).	M1
Metrics & Logs — Sub-Toggle	Top of the view: Annotator Metrics / Activity Log toggle. Annotator Metrics is the default. When Metrics is active, a Detection filter dropdown is inline, right-aligned (~1/3 width).	As a QA manager, I want to switch between metrics and audit quickly.		M1
Metrics & Logs — Annotator Metrics Table	Columns (each header has a hover InfoTip explaining the metric): Annotator, Datasets Assigned, Datasets Completed, Items Labeled, **Accuracy** ("Correct samples ÷ total reviewed samples"), **Flag**	As a QA manager, I want a single scorecard view per annotator.	Match screenshots/08_qa_metrics_logs.png. Annotator rows must reflect the live Annotator group membership.	M1

	Rate ("Open flags ÷ total items assigned"), **Label Error** ("Label corrections ÷ total resolved flags"), **Attr Error** ("Attribute corrections ÷ total resolved flags"), **Discrepancy** ("Label disagreements ÷ total overlapping items"), **Correction** ("1 – accuracy rate"). Sortable columns.			
Metrics & Logs — Activity Log Table	Chronological log of QA actions: status transitions, sample reviewed, flag resolved, dataset approved, impersonation session, etc. Columns: Action, Actor, Dataset/Detection, Details, Timestamp. Filterable by action type.	As a Super Admin, I want a single auditable timeline of every QA action.	Source is the platform audit log (req 9.1). Impersonation sessions surface as "Super Admin X impersonated Annotator Y from … to …".	M1
Finalized View	Table of datasets with status approved or finalized. Columns: Dataset name, Detection, Size, QA Status, Accuracy (if sampled), Items Labeled, Approval date. Expandable rows show QA sampling results and flag resolution stats. Export button per row (CSV/JSON).	As a downstream consumer, I want one place to find datasets that are ready to use.	Once finalized, datasets show up as available inputs in the M2 Build & Run tab.	M1
No-Lock Concurrency on Annotation	Per the prior QA architecture decision: no locking on per-image	As a team, we tolerate concurrent labeling	This matches the documented project_qa_workflow_decisions.md decisions.	M1

	annotation. Last write wins, but each label change writes an audit-log entry with actor and timestamp.	without blocking each other, and we can reconstruct the history if needed.		
Un-Finalize Allowed	A finalized dataset can be moved back to approved or needs_revision by a Super Admin. The action is logged.	As a Super Admin, I want an escape hatch when a finalized dataset turns out to need rework.	Per the prior QA architecture decision.	M1

MILESTONE 2 — Prompt Building MVP

Milestone 2 layers the core prompt-building loop on top of M1's annotated datasets. After M2, a Prompt Engineer can author a detection, draft prompt versions, run them against finalized datasets, review predictions in HiL, and monitor runs centrally — but the automated improvement and final-eval features are still deferred to M3.

Tabs in scope: **Detection Setup**, **Build & Run Datasets**, **HiL Review**, **Detections & Logs**, **Admin**.

Section 5: Detection Setup Tab (M2)

Purpose. Detection Setup is where a detection (e.g. "Major Corrosion", "Missing Siding") is defined and its prompt versions are authored, tested with quick ad-hoc images, and approved as the current baseline. Every other prompt-related tab keys off the detection picked in this tab's workspace header.

Workflow.

1. The Prompt Engineer creates a new detection or selects an existing one. The detection record captures code, display name, category (INCORRECT_CAPTURE or HAZARD_IDENTIFICATION), description, label policy, decision rubric, optional segment taxonomy, and metric thresholds.

2. They author one or more prompt versions. Prompts are structured (detection identity / label policy / decision rubric / addendum / output schema / examples) and can be scaffolded from natural language via Prompt Assist (Gemini).
3. They use Quick Test to drop in a few ad-hoc images and see live inference results before committing to a full run.
4. They approve one prompt version as the current baseline. That version is what downstream tabs default to.

Reference screenshots: Screenshot 2026-06-02 at 2.41.17 PM.png, Screenshot 2026-06-02 at 3.21.15 PM.png, Screenshot 2026-06-02 at 3.22.12 PM.png, Screenshot 2026-06-02 at 3.29.36 PM.png, Screenshot 2026-06-02 at 3.32.00 PM.png, Screenshot 2026-06-02 at 3.39.23 PM.png.

Requirement	Details	User Story	Notes	Milestone
Detection CRUD	Form to create or edit a Detection: detection_code, display_name, detection_category, description, label_policy, decision_rubric (array of bullets), segment_taxonomy (optional array), metric_thresholds (precision/recall/f1 floor used for regression pass-fail). View mode renders read-only; Edit mode renders form.	As a Prompt Engineer, I want to capture detection metadata once and reference it everywhere.	Only Super Admin can delete a detection.	M2
Workspace Header — Detection Selector	Persistent header across all workflow tabs with a Detection dropdown. Changing detection re-scopes the entire workflow.	As a user, I want to stay on the same conceptual detection as I move through tabs.		M2
Prompt Version Table	Columns: version_label, model, created_at, change_notes, approval status badge. Actions per row: View, Edit, Duplicate, Approve as Baseline, Delete (Super Admin only).	As a Prompt Engineer, I want to see all prompt iterations for a detection in one place.	Approval is mutually exclusive — approving a new version unsets the prior.	M2
Create Prompt Version	"New Prompt Version" opens a modal with two options: Start Blank or	As a Prompt Engineer, I want a fast	Model list pulled from Admin's	M2

	Use Prompt Assist. Captures version_label, model selection (Gemini family), decoding parameters (temperature, top_p, max_output_tokens), and structured prompt sections.	path to a working first draft.	allowed-models config.	
Prompt Structure Editor	The user prompt is edited as discrete sections: detection_identity, label_policy, decision_rubric, user_prompt_addendum, output_schema, examples. Each section is its own labeled editor. Reassembled into a single prompt at runtime.	As a Prompt Engineer, I want surgical control of each part of the prompt without losing structure.	Default content for each section comes from the Admin category templates (see Section 9).	M2
Prompt Assist (Gemini)	A button in the create-prompt flow that takes the detection metadata + a short natural-language brief and calls Gemini to scaffold the prompt sections. User can accept, edit, or discard.	As a Prompt Engineer, I want a head start instead of staring at a blank editor.	Uses the Prompt Assist template defined in Admin.	M2
Quick Test	Drag-drop one or many ad-hoc images, run the current prompt against them, see results in-line (decision, confidence, evidence text, raw response). Results are not persisted.	As a Prompt Engineer, I want to sanity-check a prompt before queuing a full run.	Same model + parameters as the prompt version.	M2
Approve as Baseline	"Approve" button on a prompt version sets it as the detection's current baseline. The badge updates and downstream defaults point to it.	As a Prompt Engineer, I want a clear "this is the current best" marker.		M2
Regression Result Card	If a prompt version has been tested against the detection's golden set (from M3 Held-Out Eval	As a Prompt Engineer, I want to see at a glance	Populated by M3, but the card	M2

	flow), show pass/fail vs metric_thresholds plus the actual numbers.	whether this version cleared the bar.	slot exists in M2.	

Section 6: Build & Run Datasets Tab (M2)

Purpose. Build & Run is where a Prompt Engineer either points an existing finalized dataset (from M1) at a prompt version and runs it, or — for one-off needs — builds a small ad-hoc dataset on the fly and runs it immediately. The output is a Run record with one prediction per image and a metrics summary if ground truth exists.

Workflow (Load mode).

1. Select an existing dataset for this detection.
2. Select a prompt version (defaults to the approved baseline).
3. Click Run. Watch the progress bar (live count / total with %).
4. On completion, see the metrics summary card and jump to HiL Review.

Workflow (Build mode).

1. Toggle to Build mode. Upload images (files / Excel / JSON).
2. Label each image (DETECTED / NOT_DETECTED / UNSET) and optionally apply segment tags.
3. Name the dataset, choose split type, save. (This creates the dataset just like Saved Datasets does, with the same image-ID uniqueness rules.)
4. Pick a prompt version and Run.

Reference screenshots: Screenshot 2026-06-03 at 4.26.27 PM.png, Screenshot 2026-06-03 at 4.26.41 PM.png, Screenshot 2026-06-03 at 5.11.02 PM.png.

Requirement	Details	User Story	Notes	Milestone
Mode Toggle	Load / Build toggle at the top of the tab.	As a Prompt Engineer, I want both "use an existing dataset" and "build		M2

		something quick" paths in one place.		
Load Mode — Dataset Selector	Dropdown of datasets for the current detection plus an "Unassigned" group for cross-detection datasets.	As a user, I want to find the right dataset fast.		M2
Load Mode — Prompt Selector	Dropdown of prompt versions for the current detection, with the approved baseline preselected.	As a user, I want the default to be the right answer most of the time.		M2
Run Execution	Clicking Run creates a Run record (status=running), kicks off Gemini inference per image, and streams progress. Cancel button available while running.	As a Prompt Engineer, I want to launch a run with one click and bail out if I see a problem.	Must handle 600-image runs end-to-end (req 8.2).	M2
Progress Display	Live X/Y count with percentage, current status badge, estimated time remaining (optional).	As a user, I want to know if I have time for coffee.		M2
Metrics Preview	On completion, show metrics_summary if ground truth is available: TP, FP, FN, TN, precision, recall, f1, accuracy, parse-fail rate. If no ground truth, hide metrics and show only the prediction count.	As a Prompt Engineer, I want immediate feedback on how the run went.		M2

Dataset Items Display	Below the metrics, a table of items: thumbnail, current ground-truth label, AI prediction badge, segment tags, "review flag" indicator (amber) if an annotator flagged that image. Clicking opens the Image Preview Modal with prediction details.	As a user, I want to spot-check results without leaving this tab.		M2
Build Mode — Upload Methods	Files (drag-drop), Excel/CSV (image_id, image_uri, ground_truth_label, segment_tags), JSON (structured array).	As a user, I want flexibility in how I get images in.	Image IDs must obey global uniqueness (req 4.4).	M2
Build Mode — Inline Labeling	Per-image label buttons (DETECTED / NOT_DETECTED / UNSET) and segment-tag toggles.	As a user, I want to label a small set without leaving the tab.		M2
Build Mode — Save Dataset	Captures name, split_type, optional auto-split (for MASTER). Persists like any Saved Datasets dataset.	As a user, I want my ad-hoc dataset to be reusable later.		M2
Quick Test Shortcut	Top-right "Quick Test" button mirrors the Detection Setup Quick Test — drop a handful of images, run current prompt, see results live, nothing persists.	As a user, I want a faster path than building a real dataset when I just need a smoke test.		M2
Cancel Run	While a run is running, the user can cancel; cancelled runs are persisted as status=cancelled with the partial predictions kept.	As a user, I want to abort a run that's clearly going sideways.		M2

Section 7: HiL Review Tab (M2)

Purpose. HiL Review is where a Prompt Engineer manually reviews model predictions from a Run, corrects labels, tags errors, flags ambiguous cases for secondary review, and watches metrics update live. Its output — the corrected labels and error tags — is the input to Prompt Feedback in M3.

Workflow.

1. Pick a completed Run for the current detection.
2. Filter predictions by error class (All / FP / FN / Parse Fail / Correct / Corrected / Flagged Open / Flagged Resolved).
3. Toggle between table view and image-by-image view.
4. For each item, confirm or correct the label, optionally assign an error tag, optionally leave a reviewer note, optionally flag for secondary review.
5. Watch the live metrics card update with the corrected confusion matrix.

Reference screenshots: Screenshot 2026-06-03 at 5.18.19 PM.png, Screenshot 2026-06-03 at 5.47.34 PM.png.

Requirement	Details	User Story	Notes	Milestone
Run Selector	Sticky header dropdown of completed runs for this detection (most recent first).	As a Prompt Engineer, I want to switch between runs without losing context.		M2
Filter Tab Bar	Filters: All / FP (predicted DETECTED, GT NOT_DETECTED) / FN (predicted NOT_DETECTED, GT DETECTED) / Parse Fail (raw_response unparseable) / Correct / Corrected (manually adjusted) / Flagged Open / Flagged Resolved.	As a reviewer, I want to focus on the failure modes I care about.	Counts shown in each tab.	M2
View Toggle	Table view (paginated) and Image view (carousel with	As a user, I want a		M2

	keyboard nav).	power-user view (table) and a deep-focus view (image).		
Image View — Image Panel	Same image viewer as Annotation: zoom 1×–4×, drag-pan, arrow-key nav, X/Y counter, Copy ID, show/hide metadata.	As a reviewer, I want a consistent image-handling UX.	Reuse the Annotation image viewer component.	M2
Image View — Prediction Card	Right side: prediction badge (DETECTED / NOT_DETECTED / PARSE_FAIL), confidence score, evidence text, collapsible raw model response.	As a reviewer, I want full context on what the model said and why.		M2
Image View — Ground Truth Editor	Horizontal button row (DETECTED / NOT_DETECTED / UNSET) plus an "Update ground truth on the dataset" checkbox. If checked, the correction also writes back to the source DatasetItem (otherwise the correction only lives on the Prediction record for this run).	As a reviewer, I want to fix bad ground truth without doing it twice.		M2
Image View — Error Tag Selector	Dropdown with values: MISSED_DETECTION, FALSE_POSITIVE, INFERENCE_CALL_FAILED, AMBIGUOUS_IMAGE, LABEL_POLICY_GAP, PROMPT_INSTRUCTION_GAP, SCHEMA_VIOLATION.	As a Prompt Engineer, I want to classify why the model got it wrong so we can pattern-match later.	Error tags feed into Prompt Feedback's failure-cluster analysis (M3).	M2
Image View — Reviewer Note	Free-form textarea per item. Persists to the Prediction record.	As a reviewer, I want to leave notes for myself and		M2

		the next person.		
Image View — Flag for Secondary Review	Same flag/resolve flow as Annotation. Flagging creates a ReviewFlag tied to this Prediction (and optionally the underlying DatasetItem). Resolution happens in QA's Flags Queue.	As a reviewer, I want to defer truly ambiguous calls to a second pair of eyes.	Reuses the Annotation flag modal.	M2
Live Metrics Card	Updates in real time as corrections are submitted: confusion matrix, precision, recall, f1, accuracy.	As a reviewer, I want immediate feedback on whether my corrections move the needle.		M2
Batch Operations	From the table view: "Mark all in filter as correct", bulk error-tag assignment.	As a reviewer, I want to handle obvious cases in bulk.	Confirmation modal required for any bulk write.	M2

Section 8: Detections & Logs Tab (M2)

Purpose. A read-only operational view of every detection, every prompt version, every run, and every error — for monitoring, debugging, and exports. There is no editing here; this tab is for inspection.

Reference screenshots: project root Screenshot 2026-06-02 series may include some of these views.

Requirement	Details	User Story	Notes	Milestone
Detection Accordion	Top-level list of all detections,	As a user, I want a		M2

	expandable. Per detection: display_name, category badge, created_at, summary line ("X prompts, Y runs (Y completed, Z in progress)").	directory view of the whole system.		
Prompt List per Detection	Inside an expanded detection: prompts table (version_label, model, created_at, regression result badge). Each row expandable to view the prompt structure.	As a Prompt Engineer, I want to see prompt history without context-switching to Detection Setup.		M2
Run List per Detection	Inside an expanded detection: runs table (run_id, prompt_version, dataset, model_used, status, progress, created_at). Color-coded status. Each row expandable to run details.	As a Prompt Engineer, I want to see all runs and their state.		M2
Run Details Panel	On expand: metrics summary, prompt snapshot (system + user prompt at time of run), decoding parameters, prompt_feedback_log (if M3 generated improvements), error list.	As a debugger, I want everything about a run in one place.		M2
Error List	Table of items that errored during the run: image_id, error type, error message, raw_response (first 200 chars). Filterable by error type.	As a debugger, I want to find broken images fast.		M2
Search & Filter	Search by detection_code or display_name. Filter runs by status.	As a user, I want to find a specific		M2

		detection quickly.		
Exports	Export run as CSV (predictions + errors) and error report (error types, counts, sample messages).	As a user, I want offline analysis options.		M2
Delete Actions	Super Admin only: delete a run, delete a prompt version, delete a detection. Confirmation modal each time, listing what will cascade.	As a Super Admin, I want to clean up junk runs without involving engineering.	Reinforces role matrix (Section 1).	M2

Section 9: Admin Tab (M2)

Purpose. Admin is a Super-Admin-only tab for editing the global prompt templates and feedback-generation settings that the rest of the app reads from. There is no per-detection scope here — these are platform-wide defaults.

Requirement	Details	User Story	Notes	Milestone
Workbench Templates	Two editable templates: **Prompt Assist Template** (how new detections are scaffolded from a brief) and **Prompt Feedback Template** (how M3 generates improvement suggestions from HiL corrections).	As a Super Admin, I want to tune the meta-prompts our tool uses.		M2
Category Default Templates	Per detection category (INCORRECT_CAPTURE, HAZARD_IDENTIFICATION): default System Prompt and User Prompt that pre-populate new prompt versions for detections in that category.	As a Super Admin, I want category-appropriate starting points.		M2

Image Limits for Feedback	Integer inputs: max FP, max FN, max TP, max TN, max Parse Fail images to include in the context window when generating prompt-improvement suggestions in M3.	As a Super Admin, I want to control token costs and signal balance in the Feedback flow.	Used by M3 Prompt Feedback.	M2
Edit Mode + Save	Read-only by default with an Edit button. Edit mode has monospace editors and Save/Cancel. Last-saved timestamp shown.	As a Super Admin, I want explicit edit intent to avoid fat-fingering global defaults.	All saves audit-logged.	M2
Access Restriction	Tab itself is hidden to all non-Super-Admin roles; API returns 403.	As an org, I want only one role to be able to change platform-wide defaults.		M2

MILESTONE 3 — Prompt Enhancement & Evaluation MVP

Milestone 3 closes the loop: HiL corrections from M2 feed an automated suggestion generator (Prompt Feedback), prompt versions can be compared side-by-side (Prompt Compare), and a final regression gate (Held-Out Eval) blesses a version before it ships.

Tabs in scope: **Prompt Feedback**, **Prompt Compare**, **Held-Out Eval**.

Section 10: Prompt Feedback Tab (M3)

Purpose. Take a Run that's been reviewed in HiL and ask Gemini to propose surgical edits to the prompt sections that would have prevented the observed failures. The Prompt Engineer reviews, edits, and accepts suggestions; the tab then assembles a new prompt version and optionally smoke-tests it against the golden set.

Workflow.

1. Pick a completed, HiL-reviewed run.
2. Click "Generate Suggestions". The tool sends the prompt + failure samples (subject to the image limits set in Admin) to Gemini and gets back a structured list of suggested edits.
3. Each suggestion shows old text → new text, rationale, failure cluster, priority, risk, expected metric impact, expected parse-fail impact. The user checks the boxes they want, edits the new_text inline if needed, and creates a new prompt version.
4. Optional: run the new version on the golden set and see a pass/fail badge against metric thresholds.

Requirement	Details	User Story	Notes	Milestone
Run Selector	Sticky header dropdown of completed runs (most recent first), filtered to runs that have HiL corrections.	As a user, I want to feedback on runs I've actually reviewed.		M3
Generate Suggestions	Primary button. Calls Gemini with the prompt + sampled failure images (limits from Admin). Shows a loading state.	As a Prompt Engineer, I want a smart starting point for prompt edits.	Uses Admin's Prompt Feedback Template.	M3
Suggestions Table	Columns: select checkbox, section (which prompt slot), old_text → new_text comparison, rationale, failure_cluster, priority (1–5), risk (low/med/high), expected_metric_impact, expected_parse_fail_impact. Old/new_text inline-editable. Expand row for full text.	As a reviewer, I want enough context per suggestion to accept, edit, or reject confidently.		M3
Batch Accept/Reject	Buttons to accept all selected, reject all selected.	As a user, I want to triage long lists quickly.		M3

Create New Prompt Version	After accepting suggestions: input version_label, optional change_notes, "Create & Save". The new version's prompt sections are assembled by applying the accepted edits to the source prompt.	As a Prompt Engineer, I want one click to materialize my picks as a new version.	New version appears in Detection Setup's prompt table.	M3
Feedback Log Persistence	Accepted and rejected suggestions are persisted on the source Run (prompt_feedback_log) so future visits can see history and so audit trails are intact.	As a team, we want to know which suggestions we considered and rejected.		M3
Golden Set Regression Test	Optional: "Test on Golden Set" runs the new version on the detection's golden dataset, displays previous metrics vs new metrics, and renders a pass/fail badge against metric_thresholds.	As a Prompt Engineer, I want immediate validation that my edits help.	Result also propagates to Detection Setup's regression card.	M3
Pre-Filled Suggestions on Revisit	If a run already has a prompt_feedback_log, the tab loads the prior suggestions and selections instead of regenerating.	As a user, I want to pick up where I left off.	"Regenerate" button still available.	M3

Section 11: Prompt Compare Tab (M3)

Purpose. Side-by-side metrics comparison of multiple prompt versions on the same dataset, with a disagreement drill-down for images where the prompts diverged. Used to pick the best-performing version before Held-Out Eval.

Workflow.

1. Select up to 4 prompt versions via checkboxes.
2. Pick a dataset (the dataset selector only lists datasets where all selected prompts have completed runs).
3. View the metrics comparison table and a confusion matrix per prompt.

4. Drill into the disagreement images: each shows what each prompt predicted; click for the full preview modal with all prompts' predictions side-by-side.

Requirement	Details	User Story	Notes	Milestone
Prompt Multi-Select	Checkboxes listing prompt versions for the current detection. Hard cap at 4 selections; a 5th selection is disabled with a tooltip.	As a Prompt Engineer, I want to compare a small number of candidates head-to-head.		M3
Dataset Filter	Dropdown only shows datasets where *all* currently-selected prompts have completed runs. If the user selects a 4th prompt that has no overlap, the dropdown empties and a message explains why.	As a user, I want to be prevented from selecting an impossible comparison.		M3
Metrics Comparison Table	One row per prompt. Columns: TP, FP, FN, TN, Precision, Recall, F1, Accuracy. Cell shading highlights the best in each column.	As a reviewer, I want the winner to jump out visually.		M3
Confusion Matrices	One small matrix chart per prompt, rendered in a horizontal strip beneath the table.	As a reviewer, I want to see the shape of each prompt's errors, not just the headline numbers.		M3
Disagreement Analysis	List/grid of images where the selected	As a Prompt Engineer, I want to focus		M3

	prompts disagreed. Per row: thumbnail, image_id, ground truth (if known), and the decision from each prompt.	on where prompts actually diverge.		
Multi-Prompt Image Modal	Clicking a disagreement image opens the shared Image Preview Modal extended to show, in a side panel, the prediction (decision + confidence + evidence + raw response) from every selected prompt.	As a reviewer, I want to read each prompt's reasoning on the same image.		M3
Read-Only	This tab does not write anything (no corrections, no new prompts). Pure analysis.	As a user, I want to compare without fear of side effects.		M3

Section 12: Held-Out Eval Tab (M3)

Purpose. Run a prompt version against a held-out evaluation dataset (split_type = HELD_OUT_EVAL) as the final regression gate before deployment. Read-only results — no corrections allowed, no flagging, no re-runs against held-out data without explicit override.

Workflow.

1. Pick a prompt version.
2. Pick a held-out eval dataset (selector restricted to HELD_OUT_EVAL split type only).
3. Click "Run Evaluation". Watch progress. Cancel if needed.
4. On completion, view metrics, confusion matrix, segment-level metrics, parse-fail rate, and a pass/fail badge against metric_thresholds.

5. Optionally export the results as CSV/JSON.

Requirement	Details	User Story	Notes	Milestone
Prompt Selector	Dropdown of prompt versions for the current detection.	As a user, I want to evaluate any candidate, not just the baseline.		M3
Dataset Selector — Held-Out Only	Dropdown restricted to datasets with split_type = HELD_OUT_EVAL. Other split types are not selectable and an inline message explains why.	As an org, we want to enforce that held-out data stays held-out.		M3
Run Execution	Same Run pipeline as Build & Run, but with an allow_eval_run=true flag and write-protected predictions (no HiL corrections allowed against this Run).	As a Prompt Engineer, I want to use the same infra without accidentally muddying the held-out set.		M3
Progress + Cancel	Live progress bar, optional ETA, Cancel button.	As a user, I want feedback during long runs and an out if I started the wrong one.	Must support 600-image runs (req 8.2).	M3
Results Display	Metrics summary (TP/FP/FN/TN, precision, recall, f1, accuracy), confusion matrix chart, segment-level metrics (if segment_taxonomy exists), parse-fail rate.	As a stakeholder, I want one screen that tells me "ship it or don't".		M3
Threshold Pass/Fail Badge	Compare results to detection's metric_thresholds; render a pass or fail badge prominently.	As a reviewer, I want a clear go/no-go signal.	Result also surfaces on Detection Setup's	M3

			prompt version row.	
Export Results	CSV / JSON download of the predictions and metrics.	As an analyst, I want offline records of the eval.		M3
History	Read-only table of prior eval runs on the selected dataset: prompt version, date, metrics, pass/fail.	As a team, we want a paper trail of every regression gate.		M3
No Corrections	The HiL Review tab does not list eval runs. Predictions on eval runs are immutable.	As an org, we want held-out data to remain a true test.		M3

Appendix A — Open Questions for Engineering

These are decisions intentionally left open for the implementation team. Each should be resolved during the M1 kickoff.

1. **Concurrency policy per entity.** Recommended split: optimistic locking on Detection / PromptVersion / Dataset metadata; last-write-wins on per-image labels and notes. Confirm.
2. **Parent-child dataset cascade.** When deleting a parent dataset that has linked children, refuse or cascade? Recommendation: refuse, force explicit unlink.
3. **Signed-URL strategy when sources are already Metabase signed URLs.** Confirm with Yong whether we re-sign or pass through.
4. **Rate-limit budget for the Gemini endpoint.** Must support a single 600-image sequential run; confirm per-user and per-org caps above that.
5. **Audit-log retention.** Confirm retention policy for QA activity log and impersonation records — recommend 1 year minimum.
6. **Annotator group authoritative source.** Confirm whether platform group membership is the only source, or whether Detection Lab also maintains a local allow-list for ad-hoc external annotators.
7. **Image-ID uniqueness scope.** Confirm uniqueness is per environment (dev / staging / prod isolated) vs truly global.